Design, Implementation and Evaluation of a WebXR-Based Immersive Classroom for Engineering Education

Aarati Mahato
Bangor University
Student ID: 500786680

***Abstract*—Extended Reality (XR) technologies are increasingly being used in education to create immersive and interactive learning environments. This paper presents the design, implementation, and evaluation of a browser-based WebXR immersive classroom developed for engineering education. The system is intended to teach students the fundamentals of electricity, circuit components, current flow, and the distinction between series and parallel circuits. The application is implemented using WebXR technologies and A-Frame, and is deployed through GitHub Pages to enable easy browser-based access without additional installation. The system combines a structured virtual learning space, interactive 3D models, and instructional media. To assess its effectiveness, a mixed-method evaluation approach is adopted, including usability testing, learning assessment, and technical evaluation. The results demonstrate that WebXR-based applications can improve accessibility, engagement, and the overall learning experience in engineering education.**

***Index Terms*—WebXR, Virtual Reality, Immersive Learning, Engineering Education, Browser-Based XR, A-Frame**

I. Introduction

Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) technologies have gained increasing importance in the field of education as tools for immersive learning. Compared with traditional teaching methods that rely on slides, text, and static images, immersive technologies provide a stronger sense of presence and allow learners to interact with information in three-dimensional space. This can support engagement, comprehension, and active exploration.

This project is based on the immersive classroom scenario in which a technical topic is delivered through a browser-based virtual learning environment. Virtual reality allows users to experience and interact with computer-generated environments through integrated visual and interactive elements. In this project, WebXR was selected as the core delivery method because it enables immersive learning through standard web browsers without requiring complicated installation procedures.

The main purpose of the system is to teach engineering concepts related to electricity, current flow, and circuit structures using an accessible browser-based immersive classroom. The project also demonstrates how WebXR can be used to develop lightweight and practical educational applications.

II. Related Work

The use of immersive technologies in education has been widely explored in recent years. Previous work shows that VR and WebXR systems can improve learner engagement, support visualization of abstract concepts, and create more interactive learning experiences. Web-based XR is especially valuable because it reduces technical barriers and broadens access.

Related studies also emphasize the importance of accessibility, usability, and lightweight technical design in immersive systems. These themes influenced the design of the present project. In particular, the application was designed to be simple to access, easy to deploy, and suitable for interaction through commonly available devices and browsers.

III. System Implementation

A. System Overview

The system is a WebXR-based virtual classroom designed to teach fundamental engineering concepts through browser-based immersive interaction. Users can access the application through a standard web browser using desktop systems or VR-compatible devices. The overall design combines visual learning content, interactive 3D objects, and a structured virtual learning space.

B. Aim and Objectives

The main aim of the project is to demonstrate the potential of a browser-based virtual reality classroom in engineering education.

The objectives of the project are as follows:

- To design a structured immersive classroom environment.
- To implement interactive learning elements using WebXR technologies.
- To evaluate the effectiveness of the system within an educational context.
- To understand how the system functions technically and visually.
- To develop a deployable browser-based educational XR application.

C. Technology Stack

The system was developed using a small and accessible technology stack. A-Frame was used to create the interactive 3D environment because of its simplicity and strong WebXR compatibility. HTML and JavaScript were used to implement interaction controls and system logic. GitHub Pages was used to deploy the system online, making the application accessible through a live web link.

D. System Architecture

Figure 1 presents the system architecture used in the project. The architecture shows the relationship between the user device, web browser, WebXR and A-Frame application layer, 3D scene rendering, interaction controls, learning content, and hosted web application. This reflects the browser-based structure described in the report. :contentReference[oaicite:2]index=2

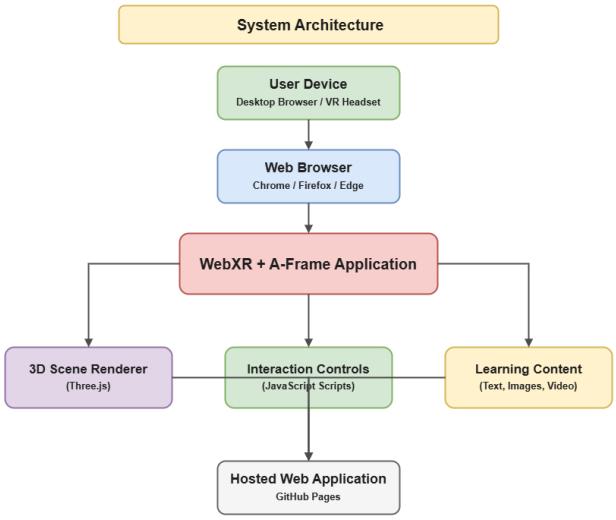

Fig. 1. System architecture design.

E. Interactive Features

The key features implemented in the application include:

- Detection of images and visual markers
- Interactive 3D models
- Real-time rendering in a browser environment

These features support engagement and active learning by allowing users to interact directly with virtual engineering content. :contentReference[oaicite:3]index=3

F. Deployment

The application was deployed through GitHub Pages, allowing the finished project to be hosted online and accessed through a standard browser. This simplified the delivery process and removed the need for local installation. The project repository link reported in the assignment is:

https://github.com/AaratiMahato/Assignment-III.git

G. Implementation Screenshots

The original report includes a long sequence of implementation screenshots showing the code structure and development environment across multiple pages. These are reproduced below as individual figures so that your final paper follows the same evidence structure as the PDF. The figure ordering matches the report's table of figures. :contentReference[oaicite:4]index=4

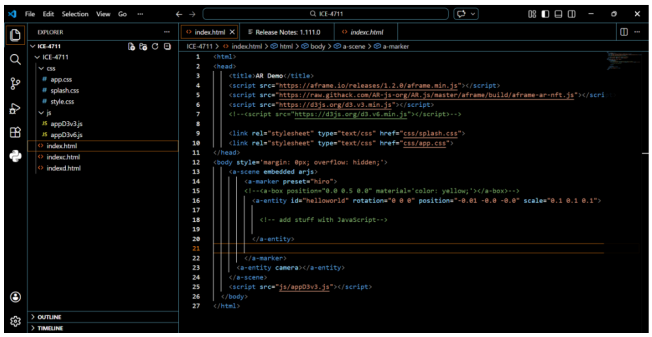

Fig. 2. System code -1.

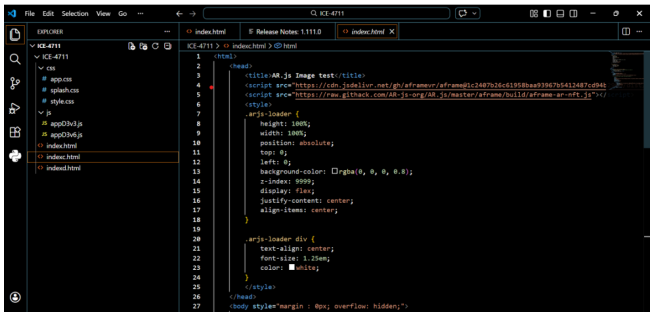

Fig. 3. System code -2.

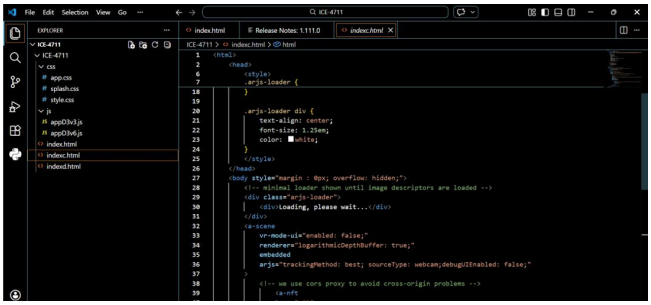

Fig. 4. System code -3.

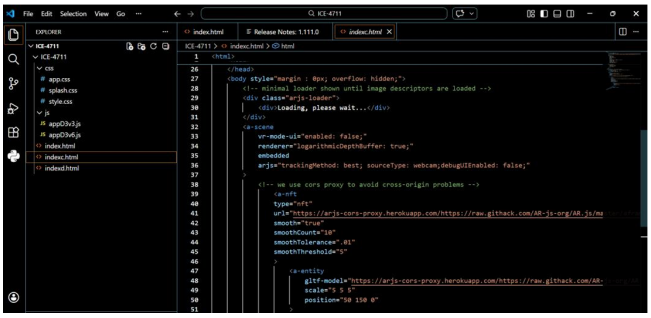

Fig. 5. System code -4.

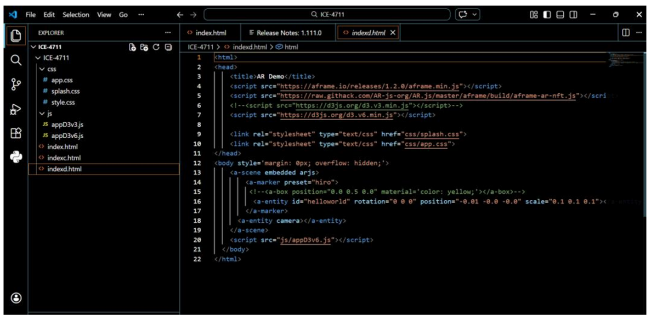

Fig. 6. System code -5.

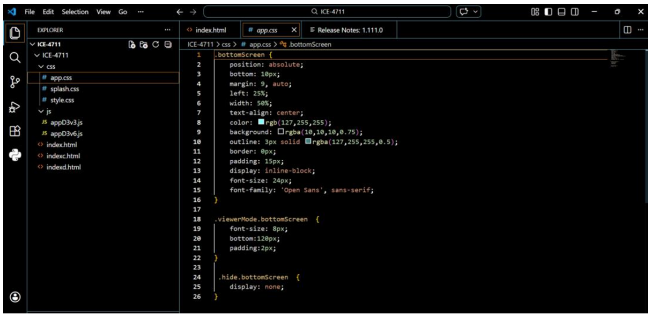

Fig. 7. System code -6.

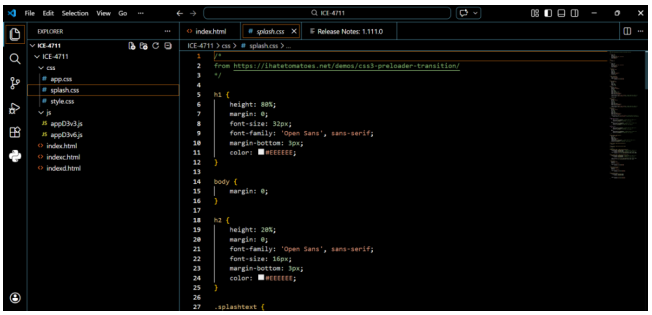

Fig. 8. System code -7.

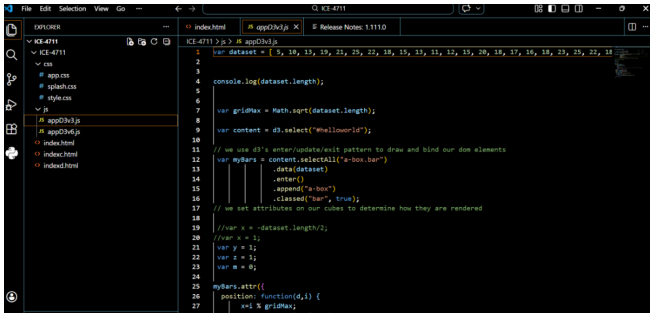

Fig. 9. System code -8.

Fig. 10. System code -9.

Fig. 11. System code -10.

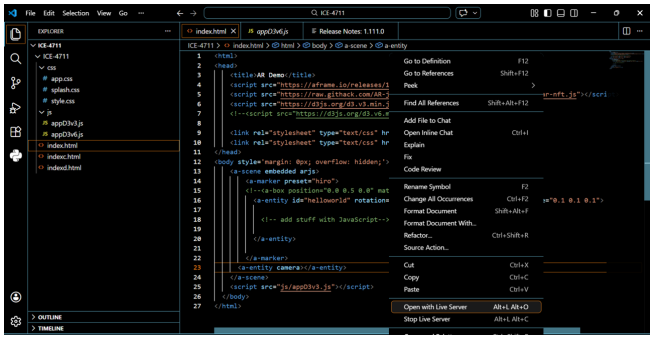

Fig. 12. System code -11.

H. System Demonstration

The report also includes demonstration screenshots showing the application in use, including marker-based interaction and the Hiro marker used by the system. These screenshots provide evidence of the functioning prototype and the output

generated when the system is tested visually. :contentReference[oaicite:5]index=5

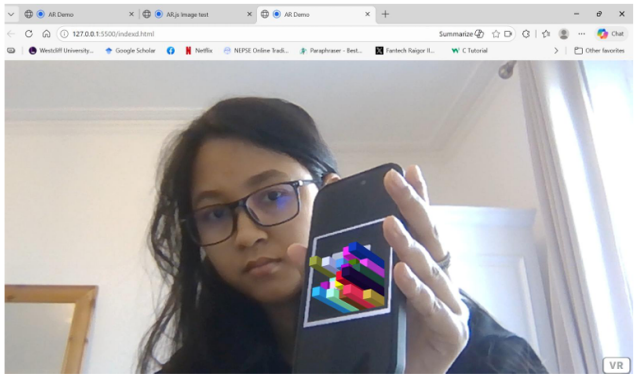

Fig. 13. System demonstration -1.

Fig. 14. System demonstration -2.

IV. Evaluation

A structured evaluation approach was proposed in the original assignment to assess the effectiveness of the immersive classroom. The evaluation includes usability testing, learning assessment, and technical evaluation. :contentReference[oaicite:6]index=6

A. Usability Testing

Usability testing was used to confirm that the system performs as expected during launch and marker-based interaction. In the first test, the browser and camera were expected to open successfully when the file was launched using Live Server. In the second test, a 3D model was expected to appear when the Hiro image was shown to the camera. Both tests were reported as successful in the assignment. :contentReference[oaicite:7]index=7

TABLE I
Usability Test 1

Action	Expected Result	Actual Result	Conclusion
First went to the code section in Visual Studio Code and then clicked open with live server in index.html	The browser should open first followed by camera opening.	The browser and camera were opened simultaneously.	The test was successfully carried out.

B. Learning Assessment

To measure learning outcomes, a simple pre-test and post-test method can be used. Learners answer questions before and after using the system so that any improvement in understanding can be observed. This approach is suitable because the immersive classroom is designed to support conceptual understanding of engineering topics such as current flow and circuit structure. :contentReference[oaicite:8]index=8

C. Technical Evaluation

Technical evaluation focuses on compatibility and performance across browsers and devices. Since the system is based on WebXR, browser support and rendering consistency are important. Testing across multiple browsers and devices helps confirm that the application remains functional and accessible in practical use. :contentReference[oaicite:9]index=9

V. Discussion

The implementation demonstrates that WebXR can support educational experiences that are more engaging than conventional teaching methods. The browser-based delivery model makes the system easier to access and reduces technical barriers for learners. The inclusion of interactive 3D models and visual testing evidence supports the value of immersive learning for engineering education.

However, the project also has limitations. The current interaction complexity is relatively basic, and the educational experience could be improved through richer content, more advanced user controls, and broader cross-device optimization. Even so, the system provides a strong foundation for future development.

VI. Conclusion

This project demonstrates the design, implementation, and evaluation of a WebXR-based immersive classroom for engineering education. The browser-based approach supports cross-platform accessibility and provides a practical way to

deliver immersive content without requiring complex installation. The evaluation strategy shows that the system is suitable for usability testing, learning assessment, and technical review. Although the current application is simple, it illustrates the wider educational potential of WebXR and immersive technology in teaching abstract engineering concepts.

REFERENCES

[1] GeeksforGeeks, "Virtual Reality Introduction," Feb. 2026. [Online]. Available: https://www.geeksforgeeks.org/techtips/virtual-reality-introduction/

[2] Tutorialspoint, "System Analysis and Design - Overview," 2026. [Online]. Available: https://www.tutorialspoint.com/system_analysis_and_design/system_analysis_and_design_overview.htm

[3] MongoDB, Inc., "What is a Tech Stack and How Do They Work?," 2026. [Online]. Available: https://www.mongodb.com/resources/basics/technology-stack

[4] Mixpanel, "What is a tech stack?," 2024. [Online]. Available: https://mixpanel.com/blog/what-is-a-technology-stack/

[5] V. Tiwari, "Understanding System Architecture: A Beginner's Guide," Feb. 2025. [Online]. Available: https://medium.com/@vaibhavtiwari.945/understanding-system-architecture-a-beginners-guide-d2aad48f0007

[6] GeeksforGeeks, "Deployment in Web Development: An Essential Guide," Jul. 2025. [Online]. Available: https://www.geeksforgeeks.org/websites-apps/deployment-in-web-development/

[7] Coursera Inc., "What Is GitHub? Use Cases and FAQ," Sep. 2025. [Online]. Available: https://www.coursera.org/articles/what-is-github

[8] M. Emmerich, "Evolution Strategies," 2018. [Online]. Available: https://www.researchgate.net/publication/332108895_Evolution_strategies

[9] IBM, "What is software testing?," 2026. [Online]. Available: https://www.ibm.com/think/topics/software-testing

[10] GeeksforGeeks, "What is Software Testing?," Mar. 2026. [Online]. Available: https://www.geeksforgeeks.org/software-testing/software-testing-basics/

[11] TendersAlerts, "Technical Evaluation," 2026. [Online]. Available: https://tendersalerts.com/en/glossary/technical-evaluation